Sanjay Dangi

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EDUCATION

Master of Science in Mechanical Engineering Aug 2023 – Dec 2025
Northern Illinois University (NIU), Dekalb, IL | GPA: 3.91/4.00
Thesis: *Characterization of a Large-Scale Electrohydrodynamic Vortex Confinement Flow in a Wire-To-Plate ESP for Particle Agglomeration.*

Bachelor of Science in Mechanical Engineering Mar 2014 – Aug 2018
Tribhuvan University, IOE Pulchowk Campus, Nepal | Percentage: 75/100
Thesis: *Design and Fabrication of Engine Powered, Manually Operated Rice Reaper Machine.*

EXPERIENCES

Graduate Research Assistant, NIU Aug 2023 – Jul 2024

- Conducted research on electrohydrodynamics (EHD) flow to induce large-scale EHD vortexes for sub-micron particle confinement and agglomeration by non-thermal plasma (NTP) effect.
- Designed and fabricated a vortex generator integrated with ESP using SolidWorks and machining process to analyze flow dynamics in electrically energized channel.
- Utilized 2D-2C particle image velocimetry (PIV) to visualize the vortex flow field seeded with 1 μm DEHS tracer particles illuminated by 1 mm thin Nd:YAG laser beam.
- Developed protocols for ESP operation, PIV/ PTV measurements and controlled particle sampling using PM sensors to assess particle agglomeration with high measurement accuracy.
- Used PIVview to extract velocity and vorticity flow fields, and performed MATLAB-based quantification of complex datasets for hypothesis validation.
- Successfully demonstrated and quantified sub-micron particle entrapment and subsequent agglomeration by large-scale EHD vortexes through real-time data curation.
- Podium presentation of preliminary results at the 42nd AAAR annual conference in Albuquerque, NM, showcasing advancements in electro-hydro dynamic flow for sub-micron particle agglomeration.

Graduate Teaching Assistant, NIU Aug 2024 – Dec 2025

- Supported undergraduate **Fluid Mechanics (MEE–340)** instruction by clarifying core concepts such as dimensional analysis, conservation laws, and internal/external flow behavior.
- Guided students in numerical problem solving, understanding governing equations, and applying fluid mechanics concepts to engineering problems.
- Assisted in teaching **Engineering Graphics (MEE–270)**, mentoring undergraduate students in SolidWorks labs and reinforcing CAD, engineering drawing, and product development concepts.
- Promoted student engagement through effective communication and hands-on exercises, reinforcing core engineering design principles.
- Managed cross-functional accreditation-related tasks and coordinated course evaluations to maintain academic standards and improve learning outcomes.

CFD and Heat Transfer Simulation Project, NIU Nov 2024 – Dec 2024
Title: *Comparative Thermal Response and Surface Temperature Analysis of Alumina Substrates Subjected to Diverse Laser Heating Modes.*

- Simulated thermal effects of stationary, moving, and pulsed lasers at multiple energy levels on various locations of an alumina substrate using **COMSOL Multiphysics** to optimize a controlled laser machining process.
- Generated time-dependent heat dissipation to visualize and analyze the heat transfer processes on alumina.
- Determined that lower-energy pulsed lasers yield uniform temperature distribution, making them ideal for additive manufacturing processes.

Title: *Simulation of Flow Around an Obstacle (MATLAB)*

- Built a full Navier–Stokes solver for 2D laminar flow using finite-difference discretization.
- Implemented pressure correction with SOR, temporary velocity prediction, and obstacle boundary enforcement.
- Calculated drag coefficient (1.53) and transient force evolution on a bluff body.
- Extended the solver to include thermal convection–diffusion and computed transient heat flux.

Title: *Heat-Driven Cavity Flow Simulation (Python and ANSYS Fluent)*

- Developed a physics-based CFD solver for natural convection in a 2D cavity using Python
- Implemented Navier–Stokes, continuity, and energy equations with Boussinesq approximation
- Applied finite difference discretization and iterative pressure–velocity coupling
- Modeled buoyancy-driven flow and temperature fields under specified boundary conditions
- Visualized velocity vectors, streamlines, and temperature contours

CAD Modeling Project, NIU Feb 2024 – Mar 2024

- Developed detailed CAD models, assemblies, and 2D technical drawings for a hydroformer (wrapping) device using SolidWorks, incorporating critical design elements such as Bill of Materials (BOM), thread notes, assembly, tolerances, and GD&T to ensure precision and manufacturability.
- Performed Finite Element Analysis (FEA) using ANSYS Mechanical to evaluate stress distributions, assess failure modes, and optimize engineering design specifications.
- Followed detail-oriented engineering standards and design guidelines, integrating mechanical design principles, DFMEA, material selection, and manufacturing considerations to enhance performance.

Fuel Cell Design & Analysis, NIU Sep 2023 – Dec 2024

- Learned electrochemical reaction kinetics and hydrogen production via electrolysis, gaining knowledge in PEM fuel cell mechanisms and water management strategies.
- Conducted critical temperature analysis to prevent membrane dry-out, optimizing heat management and enhancing fuel cell performance and reliability using Python.

Mechanical Engineering, Khilung Kalika Bio Gas Power Plant May 2019 – June 2022

- Worked with biogas based power generation systems, gaining system-level understanding of anaerobic digestion, gas handling, and energy conversion processes.
- Observed and supported operation and maintenance of mechanical components including digesters, piping networks, blowers, and power generation units.
- Applied fundamental principles of thermodynamics and fluid mechanics to interpret real plant operation, including gas flow behavior, pressure losses, and energy efficiency.
- Developed practical insight into renewable energy systems by linking theoretical coursework with real-world mechanical system performance and operational constraints.
- Handled operation and maintenance of mechanical systems across the power generation unit and bio-fertilizer (granular and liquid) production line.
- Developed insight into circular renewable energy systems by linking waste-to-energy and byproduct utilization with real-world mechanical system performance and operational constraints.

Mechanical Engineering Interns

Toyota May 2018 – Aug 2018

- Studied mechanical systems used in automotive manufacturing, focusing on equipment operation, material flow, and process efficiency.
- Applied concepts from mechanics and manufacturing courses to interpret real-time production challenges and operational constraints.
- Gained foundational knowledge of Lean Manufacturing principles through observation of standardized work, process optimization, and quality control practices.

Chaudhary Groups (CG) Foods Jan 2018 – April 2018

- Gained exposure to food manufacturing operations, developing system-level understanding of production lines, mechanical equipment, and process flow.
- Observed operation and maintenance of mechanical systems including conveyors, mixers, boilers, and packaging units under continuous production conditions.
- Related principles of thermodynamics, heat transfer, and fluid flow to real manufacturing processes such as cooking, drying, and material transport.
- Developed insight into industrial process efficiency, safety, and reliability by connecting theoretical coursework.

SKILLS

Design & CAD
SolidWorks, AutoCAD, Creo, Product design, Drafting, BOM, GD&T, Design for cost, Design for manufacturing
Simulation & Analysis Tools
ANSYS Workbench, COMSOL Multiphysics, LAMMPS, CNC Simulator
Manufacturing & Fabrication
Lathe, Water jet cutting, Molding, Welding, CAM, Lean manufacturing, Additive manufacturing, 3D printing
Mechanical Components & Systems
Gears, Bearings, Pumps, Valves, Pressure Sensors, Mechanical Assemblies, Boilers
Programming & Technical Computing
Python, MATLAB, LaTeX, GitHub
Process Improvement & Quality
DMAIC
Electronics & Instrumentation
PLC (ABB), SCADA, Arduino, PM sensor, Laser, High DC voltage
Office & Documentation Tools
MS Word, MS PowerPoint, MS Excel, Office 365

REFERENCES

Dr. Eric Monsu Lee, Assistant Professor, NIU
Supervisor (815-753-1288, elee3@niu.edu)

Dr. Kyu Taek Cho, Associate Professor, NIU
Thesis Committee Member (815-753-3346, kcho@niu.edu)

Dr. Tariq Shamim, Department Chair, NIU
Thesis Committee Member (815-753-9964, tshamim@niu.edu)